Assignment1 Micro Economics B

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1 Straffespark

1.1

Because players choose their actions simultaniously without knowledge of the other persons strategy, and the game is palyed without repetition. What makes it zero Sum - For every possible ooutcome f.ex (L,L) or (R,L) the payoff sum is constant, here 0. Here if the Goalie wins the striker looses and vice versa. We defnie the utility funciton of player 2 $\quad U_2 : U_2 = -U_1$

1.2

- Players: $P_1 = Striker, P_2 = Goalie$
- Action sets: $P_1 \in \{L, R\}$, $P_2 \in \{L, R\}$
- Payoffs:

	L	R
L	$(0.65, -0.65)$	$(0.88, -0.88)$
R	$(0.83, -0.83)$	$(0.56, -0.56)$

Best Response Function

$$BR_1 = \begin{cases} L, & \text{if } q < 0.64 \\ R, & \text{if } q > 0.64 \\ \{L, R\}, & \text{if } q = 0.64 \end{cases}$$

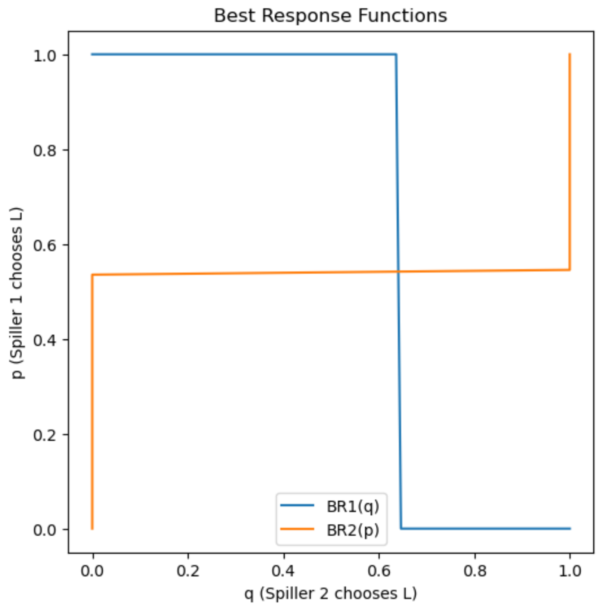

Figure 1: Best Response Functions

$$BR_2 = \begin{cases} L, & \text{if } p < 0.54 \\ R, & \text{if } p > 0.54 \\ \{L, R\}, & \text{if } p = 0.54 \end{cases}$$

All NE No pure NE, can be found by iterating best response by hand but we find a mixed NE when both players mix.

$MSNE = \{(\sigma_1, \sigma_2)\} = \{((0.64, 0.36), (0.54, 0.46))\}$

Best response functions plot

1.3

To construct a comaprison talbe for the empirical frequensies I remove the C columns from talbe 1 in the assignment and then divide by the new total nr actions 372 for each cell. I then insert the columns alongside our previous result.

Action	Kicker		Keeper	
	NE	Empirical	NE	Empirical
L	0.54	0.54	0.64	0.57
R	0.46	0.46	0.36	0.43

Table 1: Mixed-strategy Nash equilibrium vs. observed play (Chiappori et al., 2003; C action removed).

Comparing our NE to the empirical frequences we find that they are similar especially for the Kicker but that the keeper may overweight going left slightly.

1.4

Normal form 3x3 game

- Players: $\{P_1, P_2\} = \{Kicker, Keeper\}$
- Pure strategies: $S_1 = \{L, C, R\}$, $S_2 = \{L, C, R\}$
- Payoff Functions:

Kicker \ Keeper	**L**	**C**	**R**
L	(0.65, -0.65)	(1.00, -1.00)	(0.88, -0.88)
C	(1.00, -1.00)	(0.00, -0.00)	(0.89, -0.89)
R	(0.83, -0.83)	(1.00, -1.00)	(0.56, -0.56)

Table 2: Payoff matrix for kicker (Player 1) and keeper (Player 2)

Remove Dominated Strategies

There are no dominated strategies to remove with IESDS

Solving for all NE

Support enumeration finds a mixed strategy nash equilibirum as follows:

Player	L	C	R
Kicker (p^*)	0.476	0.218	0.307
Keeper (q^*)	0.521	0.185	0.294

Table 3: Mixed strategy Nash equilibrium: probabilities of playing Left, Center, and Right

Comparison with Empirical Frequencies Like before we normalise the frequences matrix from table 1 to find them as percentages to be compared with our NE f.ex we take the first row from talbe 1 divided by total nr actions, $(117 + 4 + 85)/459 = 0.449$ to find the first value:

Player	L	C	R
Kicker (empirical)	0.449	0.172	0.379
Keeper (empirical)	0.566	0.024	0.410

Table 4: Empirical action frequencies based on full data (including center)

The empirical frequencies from Chiappori et al. (2003) show that kickers choose (L, C, R) with approximate probabilities (0.45, 0.17, 0.38), while goalkeepers choose (L, C, R) with (0.57, 0.02, 0.41).

In the computed Nash equilibrium, the mixed strategies are:

$$p^* = [0.476,\ 0.218,\ 0.307] \quad \text{(Kicker)}$$
$$q^* = [0.521,\ 0.185,\ 0.294] \quad \text{(Goalkeeper)}$$

The kicker chooses Center slightly more often in equilibrium than in practice, while the goalkeeper almost never chooses Center empirically.

1.5

What I recommend Although there may be complicated reasons as of why NE is not empirically played, f.ex if one player expects the other to act irrationally. However on our analysis I would recommend trying moving towards NE to see if this imporves the results.

- Kicker should go more C,L

- Keeper should go mostly go more C over all else.

1.6

Reflection of why NE is not played

Both generally play center less than NE, this is probably a bias from the shooter to shot a place where the keeper is not already. Further in the case where the goalie goes f.ex left, placing the ball right has the same effect as sending it center ie it is on the opposite side of the direction the goalie is going in, but it is further from the goalie hence the percived risk can be smaller by just putting it as far away from him as possible. Could also be bias becuase a player is either right or left footed (never center footed) I can also be that the keeper is percieved as worse if he just stands in the middle and the ball goes in than if he went either way.

In whos interest is it to not play NE

Technically is it not in any players best interest to not play NE since by definitions we can not have any possitive deviations. It can be that there are some outside interests f.ex if the keeper is percived to be doing a worse job by just standing in the middle than going to either side this might skew the payoff function in a way not accurately represented by the underlaying data, and hence this could be a possible scource of percived deviation although this is a fault in the payoff definition.

1.7

Normal form with new data

- Players: $\{P_1, P_2\} = \{Kicker, Keeper\}$
- Pure strategies: $S_1 = \{L, C, R\}$, $S_2 = \{L, C, R\}$
- Payoff Functions:

Kicker \ Keeper	**L**	**C**	**R**
L	(0.63, -0.63)	(1.00, -1.00)	(0.94, -0.94)
C	(0.81, -0.81)	(0.00, -0.00)	(0.89, -0.89)
R	(0.90, -0.90)	(1.00, -1.00)	(0.44, -0.44)

Table 5: Payoff matrix for kicker (Player 1) and keeper (Player 2)

Finding dominated strategies IESDS found no strictly dominated strategies

Finding NE With support enumeration we find the following mixed strategy nash equilibirum:

Player	**L**	**C**	**R**
Kicker (p)	0.429	0.238	0.334
Keeper (q)	0.595	0.092	0.313

Table 6: Mixed strategy equilibrium: estimated probabilities of playing Left, Center, and Right

Comaprison with empirical frequencies We combine empirical frequencies our previous NE, and the NE calculated based on the new data into one table for easy comparison.

Player / Source	**L**	**C**	**R**
Kicker (empirical)	0.449	0.172	0.379
Kicker (Nash old) p^*	0.476	0.218	0.307
Kicker (Nash new) p	0.429	0.238	0.334
Keeper (empirical)	0.566	0.024	0.410
Keeper (Nash old) q^*	0.521	0.185	0.294
Keeper (Nash new) q	0.595	0.092	0.313

Table 7: Comparison of empirical frequencies, old Nash equilibrium, and new Nash equilibrium strategies

Based on our analysis we may recommend the following:

- Kicker: L is ok between the values of our two analysis, C is consistently undervalued, R is slightly over valued. The overall recommendation stands to go more more C, but recoomendation to go more L falls.

- Similarly for the keeper empirical L lies between the NE values of our analysis, C is consistently under valued, R is over valued. Recommendation to go more C stands.

2 Hearthstone

2.1

Zero sum game with two persons I create a zero sum game by assuing $U_2 = -U_1$

Find all strictly dominated strategies I run IESDS on the resulting utility matrixes, i find the following strictly dominated strategies: Aggro Shaman, Control Warrior, Miracle Rogue, Other Mage, Ping Mage, Rally Priest, Spell Damage Mage, Token Druid.

2.2

Finding the k'th order strategy

Order	Strategy	Row Index
1st	Secret Paladin	15
2nd	Rush Warrior	13
3rd	Control Priest	2
4th	Control Warlock	3
5th	Face Hunter	5
6th	Secret Libram Paladin	14
7th	Control Warlock	3
8th	Face Hunter	5
9th	Secret Libram Paladin	14
10th	Control Warlock	3
11th	Face Hunter	5
12th	Secret Libram Paladin	14
13th	Control Warlock	3
14th	Face Hunter	5
15th	Secret Libram Paladin	14
16th	Control Warlock	3
17th	Face Hunter	5
18th	Secret Libram Paladin	14
19th	Control Warlock	3
20th	Face Hunter	5

Table 8: Cognitive hierarchy strategies from 1st- to 20th-order

We observe that after k=3 we get a repeating action sequence of 3,5,14. Control Warlock, Face Hunter and Secrete Libram Paladin. Meaning the best response sequence eventually loops back to the original action, as k increases hence creating a loop.

2.3

Choice of deck based on strategy shares Intuitively it seems like choisng the deck that is most likely to win against the most picked deck seems like a viable strategy, looking at the heat map we would select that which is most effecitve against NoMinionMage, which is Face hunter with a 67.66 % win rate.

Taking things a bit further we want to take into account all the choises our oponent can make, maybe there is some deck that works against all other decks except for the most picked one that overall would be better, we can build a simple simulation. We find that through 1000 simulatons we still get Face hunter having the highest win rate given our selection distribiton for the opponent. So Arne should play FaceHunter: TODO: check if wrong. (right secret paladin?)

2.4

NE and MNE We use linear programming to find our mixed NE at Control Warlock: 24%, Rush warrior: 71%, Secret Paladin 5% .

Argument for choise of solution algorithm: To find the mixed-strategy Nash equilibrium, we chose to use linear programming, as it is an effective method for solving two-player zero-sum games with a large number of strategies. Specifically, we use the method to maximize the player's minimum gain (minimax), which ensures that the player selects a strategy that protects against the worst possible outcome. This method is suitable because the payoff matrix for the Hearthstone game is known and fixed, and both players are assumed to choose their strategies simultaneously. Alternatives such as support enumeration would be impractical due to the number of strategies (20+), which makes linear programming a more scalable and accurate choice.

2.5

We find when running it fictitious play 1000 times that the results vary of opposed to when running the algorithm 100.000 times, where the results seem more stalbe indicating convergence.

Deck	**Fictitious Play (1k)**	**Fictitious Play (100k)**	NE
Control Priest	0.002	0.000	0.00
Control Warlock	0.143	0.241	0.24
Rush Warrior	0.790	0.710	0.71
Secret Paladin	0.065	0.049	0.05

Table 9: Fictitious play probabilities after 1k and 100k iterations

We observe in the talbe that 1000 ittereations is not enough for full convergence to our NE at Control Warlock: 24,2 % , Rush Warrior: 70,9 % , Secret Paladin: 4.9 % . We estimate that we have convergence after 100k itterations, at least to a two point decimal percition.

When do we have convergence? It looks like we have some built in oscillations in the algorithm, we do get lower variance for higher nr itterations but it is hard to tell exactly when we converge since it is not smooth convergence. We would have to define convergenece in terms of some threshold ϵ of acceptable persition. However 100k iiterations gives us a very similar result to our NE.

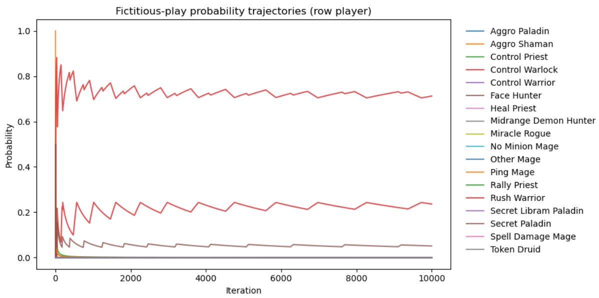

Figure 2: Caption

3 Priskonkurrence i Bilmarkedet

3.1.1

We assume that both firms choose prices simultaneously and maximize their profit given the price of the other firm. For each price level of the competitor, there exists a price level that maximizes the firm's profit, this is called the best-response function.
We solve the Nash equilibrium numerically by finding the intersection point of the two best-response functions. The graph below illustrates these functions.

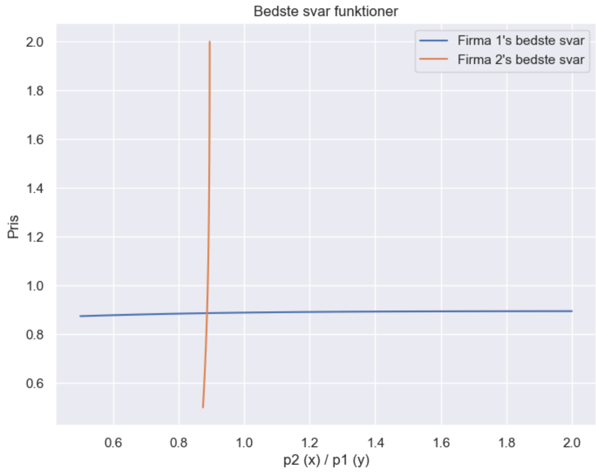

Figure 3: The best-response functions of the two firms. The intersection point represents the Nash equilibrium

3.1.2

We now assume that the two firms form a cartel and therefore maximize the total joint profit. This means that they jointly choose the prices p_1 and p_2 to maximize

$$\pi^{Cartel}(p_1, p_2) = \pi_1(p_1, p_2) + \pi_2(p_1, p_2)$$

The optimal solution is found numerically. The result shows that the optimal cartel price for both firms is $p_1 = p_2 = 0.941$, which is higher than the Nash equilibrium price, as expected.

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Kartelpriser: p1 = 0.941, p2 = 0.941
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Figure 4: Cartel prices, where both firms set the same price to maximize joint profit

3.1.3

In the Stackelberg equilibrium, we assume that Firm 1 is the leader and chooses its price first. Firm 2 observes this and responds optimally (chooses its best-response function). The solution is found by:

1. Expressing Firm 2's best-response function given p_1
2. Substituting this function into Firm 1's profit and maximizing with respect to p_1

The result shows that the leader firm sets a slightly higher price than the follower firm. The Stackelberg equilibrium is close to the Nash equilibrium in this case.

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Stackelberg-ligevægt:
p1 (leder) = 0.888
p2 (følger) = 0.887
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Figure 5: Stackelberg equilibrium, where Firm 1 sets the price first and Firm 2 responds optimally

3.1.4

The size of the parameter α has a significant impact on the market outcome in all three cases. α measures the price sensitivity of consumer choices, the higher the α, the more sensitive consumers are to price changes.
When α is low, consumers are relatively insensitive to prices, which allows firms to set higher prices. This results in high prices in the Nash, cartel, and Stackelberg solutions. Under cartel formation, the price is especially high because consumers are less likely to switch away in response to price increases.
When α is high, competition intensifies, as small price differences have a large effect on demand. This pushes prices down in the Nash equilibrium, and even cartels have limited ability to raise prices without losing significant market share.
Examples of markets with low α include pharmaceuticals, luxury brands, or cars with strong brand identity. In contrast, markets with high α are characterized by low differentiation and high transparency, such as low-cost airlines, groceries, and gasoline.

3.2.1

Using the Manual Iterated Best Response algorithm, the Nash equilibrium was found by allowing each firm to optimize its price given the other firm's price. Upon convergence, a price profile is reached in which neither car manufacturer has an incentive to unilaterally deviate from its strategy.

The outcome of the algorithm yielded the following Nash equilibrium prices:

- $p_1 = 50{,}000$ EUR (VW Passat)
- $p_2 = 50{,}000$ EUR (Ford Focus)

The observed prices in 1999 were:

- VW Passat: 37800 EUR
- Ford Focus: 25,850 EUR

As seen, the computed Nash prices are significantly higher than the observed prices. This discrepancy can be attributed to several factors:

1. The model includes only two cars, while in reality, there are many competitors in the market. The reduced level of competition in the model allows firms to set higher prices.
2. The estimated demand structure and parameters (e.g., price sensitivity) are based on the entire market rather than solely on Germany in 1999. This can lead to imprecise predictions for individual years.
3. It is assumed that marginal costs are 50% of the sales price, which may not accurately reflect reality.

Overall, the analysis indicates that the simple model, in which only two cars interact, overestimates the market power of the firms compared to the real world.

3.2.2

In this section, we assume that firm 1 and firm 2 form a cartel and cooperate to set prices in order to maximize total profit. The optimal pricing is determined by maximizing the sum of the firms' profits, taking into account the joint demand.

The results show the following cartel prices:

- $p_1 = 50{,}000$ EUR (VW Passat)
- $p_2 = 48{,}823.75$ EUR (Ford Focus)

Compared to the Nash equilibrium from Section 3.2.1, where both firms set prices around 50,000 EUR, the cartel prices are only marginally higher (or the same). This may be due to the fact that the model already implies limited competition (only two cars) and that consumers have relatively low price sensitivity.

Cartel Risk

In markets with few players, such as the car market in this simplified model, firms may achieve nearly cartel-like outcomes even without explicit coordination. This highlights the importance of antitrust authorities' oversight, particularly in industries with high entry barriers and low product differentiation.

The model illustrates that collusion in such markets can lead to higher prices and lower consumer welfare. However, the difference between Nash and cartel prices depends on the market structure and consumers' price elasticity.

3.2.3

This section analyzes the competition between car manufacturers VW and Opel at the brand level. Each firm selects a single common relative price factor for all their cars, based on baseline prices.
Nash equilibrium:

- $p_1 = 2.000$ (VW)
- $p_2 = 1.852$ (Opel)

Cartel solution:

- $p_1 = 2.000$
- $p_2 = 2.000$

The cartel here chooses to set prices at the maximum allowed level. This suggests that the model implies low competitive pressure and high market power for both manufacturers, both under Nash and cartel outcomes.
When the Nash equilibrium already lies at the price ceiling, the difference compared to the cartel outcome diminishes. This indicates that even without coordination, the market outcome can resemble a monopoly — especially in models with few players and low price sensitivity. In practice, such outcomes may occur in markets characterized by high entry barriers and strong brand loyalty, where firms can set high prices almost independently of one another.

3.2.4

Issue A (Relevant):
Will mergers reduce price competition and thereby delay the green transition through higher electric vehicle prices?

The results from the analysis show that when the number of players is reduced and competition between brands or car models is limited, firms' pricing power increases significantly. This is clearly reflected in both the Nash and cartel outcomes, where prices in some cases reach the maximum allowed level.

If mergers in the car market reduce the number of independent producers, this may lead to cartel-like behavior and higher prices, including for green technologies such as electric vehicles. This would make it more difficult for consumers to switch to greener modes of transport, thereby hindering the green transition. Hence, the effect of mergers on price competition is a relevant issue that the models in this analysis can directly address.

Issue B (Not Relevant):
Will mergers lead to higher investments in green technology through economies of scale and R&D synergies?

This issue concerns whether larger automotive groups, post-merger, would have better financial conditions for investing in green technology and developing, for example, electric vehicles and batteries. In principle, this could accelerate the green transition.

Our model and analyses focus solely on demand structure and pricing strategies in a static setup, without considering investment, innovation, or cost dynamics. Therefore, we cannot assess the effect of mergers on incentives for green innovation, and this issue falls outside the scope of the model.

In summary, the analysis shows that mergers in the car market may lead to higher prices and thereby delay the green transition, an effect that can be modeled and quantified. Conversely, the analysis says nothing about potential benefits of mergers in the form of increased investment or innovation. Such dynamic aspects must be analyzed within a more extended modeling framework.